

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

Granular Synthesizer Plugin | *C++*, *JUCE*, *VST3*, *AU*

github.com/Verdent06/granular-synth

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic `shared_ptr` swap so the UI thread never blocks `processBlock()`.
- Configured CMake to compile VST3 and AU bundles from a single JUCE codebase — targeting a macOS universal binary (arm64 + x86_64) for native Apple Silicon and Intel — and audited every `processBlock()` path against a real-time safety checklist confirming zero heap allocations and zero lock acquisitions before release builds.

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.

Vylet

May 2026 – Present | vyletdata.com

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three clients

- Engineered a LangSmith eval pipeline spanning 20 adversarial business test cases across 13 archetype labels — manufacturers, SaaS tools, PE holding companies, geographic mismatches — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.
- Diagnosed a name-collision defect in ownership-verification logic that was incorrectly rejecting valid acquisition targets sharing a name with an unrelated business elsewhere; the fix lifted the pipeline's lead-qualification rate from 79% to 89% with zero change to sourcing volume.

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at ~2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating ~800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.

PROJECTS

SignalWeaver | *Python*, *FastAPI*, *PostgreSQL*, *pgvector*

github.com/Verdent06/SignalWeaver

- Containerized the stack with Docker Compose (API + Postgres/pgvector + nginx frontend) and a GitHub Actions CI pipeline (frontend build, pytest, API image build on main).
- Served composite financial-research scores through async REST endpoints (FastAPI) wrapping fundamentals, MPNet sentiment, and regression logic, instrumented end-to-end at 9.1s p50 / 15.2s p99 across 90 successful runs on 90 tickers.

TECHNICAL SKILLS

Languages Python, TypeScript, C++, SQL

Frameworks JUCE, Flask, Angular, RxJS, FastAPI

Cloud & Infrastructure AWS (EC2, S3), Docker, GitHub Actions, Git